INTRODUCTION TO INFORMATION SECURITY

WEEK 5

Substitution Cipher

- Polyalphabetic (Vigenère) Cipher: this algorithm uses different cipher letters for different occurrences of same plaintext letter.

where m is the length of the key string.

Security of Vigenere Cipher:

(1) Much more secure than Caeser cipher

(2) Need to determine key size

(3) Repetitions in ciphertext give clues to period.

Vigenere Cipher

★ It consists of the 26 Caesar ciphers with shifts of 0 through 25.

Encryption process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Decryption process:

$$P_i = (C_i - K_{i \bmod m}) \bmod 26$$

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17
CT	25												

Key	15	19	8	21	4	3	4	2	4	15	19	8	21	4
PT	4	3	18	0	21	4	24	14	20	17	18	4	11	5

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17
CT	25												

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17
CT	25	8	2	21	19	22	16	13	6	17	25	6	21

Substitution Cipher

Vigenere Cipher

The encryption formula is defined as $C_i = E_k(P_i) = (P_i + K_{i \bmod m}) \bmod 26$
while the decryption formula is $P_i = D_k(C_i) = (C_i - K_{i \bmod m}) \bmod 26$
where m is the length of the key string.

Example: Let the key string be gold. Using the encoding rule a = 0, b = 1, ..., z = 25, the numerical representation of this key string is (6, 14, 11, 3)

plaintext: proceed meeting as agreed
key: goldgol dgoldgo ld goldgo ----->
ciphertext: vfzfkso pkseltu lv guchkr

p	r	o	c	e	e	d	m	e	e
15	17	14	2	4	4	3	12	4	4
g	o	l	d	g	o	l	d	g	o
6	14	11	3	6	14	11	3	6	14
v	f	z	f	k	s	o	p	k	s
21	5	25	5	10	18	14	15	10	18

Hill Cipher

- ★ Multi-letter cipher.
- ★ Developed by Lester Hill in 1929.
- ★ Encrypts a group of letters : digraph, trigraph or polygraph.

The Hill Cipher

- Plaintext is broken into blocks of size m.
- Here, *Key is a m X m matrix of integers between 0 to 25.*
- Let $p_1, p_2, p_3, \ldots, p_m$ be the numeric representation of characters in plaintext.
- Let $c_1, c_2, c_3, \ldots, c_m$ represent corresponding characters in ciphertext.
- To compute a ciphertext, we use the mapping,

A -> 0, B -> 1, . . . , Z -> 25

- Now,

$$c_1 = p_1 k_{11} + p_2 k_{21} + \ldots + p_m k_{m1} \bmod 26$$

$$c_2 = p_1 k_{12} + p_2 k_{22} + \ldots + p_m k_{m1} \bmod 26$$

$$\ldots$$

$$c_m = p_1 k_{1m} + p_2 k_{2m} + \ldots + p_m k_{mm} \bmod 26$$

In general, $\boxed{c = p\,K}$

where, i) c and p are row vectors of ciphertext & plaintext resp'ly,

ii) K is m X m matrix comprising the key

At Receiver, plaintext can be recovered by,

$$p = c\,K^{-1}$$

The Hill Algorithm

This can be expressed as

$C = E(K,P) = P \times K \bmod 26$

$P = D(K,C) = C\,K^{-1} \bmod 26 = P \times K \times K^{-1} \bmod 26$

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

Encryption

$$C_1 = (P_1K_{11} + P_2K_{21} + P_3K_{31}) \bmod 26$$
$$C_2 = (P_1K_{12} + P_2K_{22} + P_3K_{32}) \bmod 26$$
$$C_3 = (P_1K_{13} + P_2K_{23} + P_3K_{33}) \bmod 26$$

Hill Cipher (Rules for Encryption)

1. Assign a number to each character of the Plain-Text, like (a = 0, b = 1, c = 2, … z = 25). As per given table.

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
13	14	15	16	17	18	19	20	21	22	23	24	25

For Example,

Plain Text = SUNDAY

S = 18, U = 20, N = 13, D = 3, A = 0, Y = 24

2. 2x2 or 3x3 key matrix is given,

$$\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \quad \text{OR} \quad \begin{bmatrix} 10 & 12 & 7 \\ 2 & 32 & 13 \\ 22 & 1 & 91 \end{bmatrix}$$

3. Make a group of plain text as per given key matrix size. Each pair of plain text multiply with key matrix.

For Example,

Plain Text = SUNDAY

If key matrix is 2 x 2,

Plain text divided in into group of 2 alphabets : SU ND AY

If key matrix is 3 x 3,

Plain text divided in into group of 3 alphabets : SUN DAY

4. Multiplication of plain text matrix and key word matrix generate new matrix.

5. Newly generated matrix values modules with 26.

6. After modules 26, matrix values assign characters using rule no 1 table. It generates final cipher text.

Hill Cipher (Rules for Decryption)

1. Assign a number to each character of the cipher text, like (a = 0, b = 1, c = 2, … z = 25). As per given table.

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

For Example,

Cipher text = SUNDAY

S = 18, U = 20, N = 13, D = 3, A = 0, Y = 24

2. Find the inverse of given key matrix (2x2 or 3x3),

3. Make a group of Cipher text as per given key matrix size. Each pair of Cipher text multiply with inverse key matrix.

For Example,

Cipher text = SUNDAY

If key matrix is 2 x 2,

cipher text divided in into group of 2 alphabets : SU ND AY

If key matrix is 3 x 3,

cipher text divided in into group of 3 alphabets : SUN DAY

4. Multiplication of Cipher text matrix and inverse key word matrix generate new matrix.

5. Newly generated matrix values modules with 26.

6. After modules 26, matrix values assign characters using rule no 1 table. It generates final plain text.

Hill Cipher (Encryption)

Example: Plain text is CD. Find out cipher text using hill cipher. Decrypt cipher text using hill cipher.

Key Matrix = $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

Solution:

- Plain Text = CD (C = 2, D = 3)
- Key matrix = $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} C \\ D \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 13 \\ 18 \end{bmatrix} \% \ 26 \Rightarrow \begin{bmatrix} 13 \\ 18 \end{bmatrix} \Rightarrow \begin{bmatrix} N \\ S \end{bmatrix}$$

$\rightarrow ((2 * 2) + (3 * 3)) \rightarrow 4 + 9 \rightarrow 13$
$\rightarrow ((3 * 2) + (4 * 3)) \rightarrow 6 + 12 \rightarrow 18$

$\boxed{\textbf{Cipher Text = NS}}$

Hill Cipher Example

Question: Encrypt "pay more money" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Solution:

p	a	y	m	o	r	e	m	o	n	e	y
15	0	24	12	14	17	4	12	14	13	4	24

Key = 3 x 3 matrix. PT = pay mor emo ney

Hill Cipher Example

Encrypting: pay

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (15\ 0\ 24)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (15\text{x}17+0\text{x}21+24\text{x}2 \quad 15\text{x}17+0\text{x}18+24\text{x}2 \quad 15\text{x}5+0\text{x}21+24\text{x}19) \bmod 26$$

$$= (303\ 303\ 531) \bmod 26$$

$$= (17\ 17\ 11) \qquad = (R\ R\ L)$$

Hill Cipher Example

Encrypting: mor

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (12\ 14\ 17) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (12\text{x}17+14\text{x}21+17\text{x}2 \quad 12\text{x}17+14\text{x}18+17\text{x}2 \quad 12\text{x}5+14\text{x}21+17\text{x}19) \bmod 26$$

$$= (532\ 490\ 677) \bmod 26$$

$$= (12\ 22\ 1)$$

$$= (M\ W\ B)$$

Encrypting: emo

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (4\ 12\ 14)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (4x17+12x21+14x2 \quad 4x17+12x18+14x2 \quad 4x5+12x21+14x19) \bmod 26$$

$$= (348\ 312\ 538) \bmod 26$$

$$= (10\ 0\ 18)$$

Encrypting: ney

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (13\ 4\ 24)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (13\text{x}17+4\text{x}21+24\text{x}2 \quad 13\text{x}17+4\text{x}18+24\text{x}2 \quad 13\text{x}5+4\text{x}21+24\text{x}19) \bmod 26$$

$$= (348\ 312\ 538) \bmod 26$$

$$= (15\ 3\ 7)$$

$$= (P\ D\ H)$$

Hill Cipher Example

Question: Encrypt "pay more money" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B	K	A	S	P	D	H

Hill Cipher (Decryption)

1. Find out inverse matrix of given key matrix.

➔ $K^{-1} = \frac{1}{|K|} * K_{adj}$

➔ $|K| = \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 8 - 9 = -1$

➔ $K_{adj} = \begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix}$

For |k| Replace diagonal values and others value's signs then multiply diagonal values

$$\begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix}$$

=4*2 + (-3*-3) = -1

For Kadjoint Replace diagonal values and others value's signs

$$\begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix}$$

➔ $K^{-1} = \frac{1}{|K|} * K_{adj} = \frac{1}{-1} * \begin{bmatrix} 4 & -3 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix}$

Hill Cipher (Decryption)

- Cipher Text = NS (N = 13, S = 18)
- Key Inverse Matrix = $\begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix}$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$\begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix} \cdot \begin{bmatrix} N \\ S \end{bmatrix} \Rightarrow \begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix} \cdot \begin{bmatrix} 13 \\ 18 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 3 \end{bmatrix} \% \, 26 \Rightarrow \begin{bmatrix} 2 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} C \\ D \end{bmatrix}$$

$\rightarrow ((-4 * 13) + (3 * 18)) \rightarrow -52 + 54 \rightarrow 2$

$\rightarrow ((3 * 13) + (-2 * 18)) \rightarrow 39 - 36 \rightarrow 3$

Plain Text = CD

The Hill Cipher Decryption for 3x3 matrix

$$K: \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Decryption requires K^{-1}, the inverse matrix K.

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

To find Det K, Adj K

$$\mathrm{Det}\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$\mathrm{Det}\begin{pmatrix} 17 & & \\ & 18 & 21 \\ & 2 & 19 \end{pmatrix} \bmod 26 \qquad \mathrm{Det}\begin{pmatrix} & 17 & \\ 21 & & 21 \\ 2 & & 19 \end{pmatrix} \bmod 26$$

$$= 17(18\times19 - 2\times21) \qquad = 17(18\times19 - 2\times21) - 17(19\times21 - 2\times21)$$

$$\mathrm{Det}\begin{pmatrix} & & 5 \\ 21 & 18 & \\ 2 & 2 & \end{pmatrix} \bmod 26$$

$$= 17(18\times19 - 2\times21) - 17(19\times21 - 2\times21) + 5(2\times21 - 2\times18)$$

$$\text{Det}\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= 17(18\text{x}19 - 2\text{x}21) - 17(19\text{x}21 - 2\text{x}21) + 5(2\text{x}21 - 2\text{x}18) \bmod 26$$

$$= 17(342 - 42) - 17(399 - 42) + 5(42 - 36) \bmod 26$$

$$= 17(300) - 17(357) + 5(6) \bmod 26$$

$$= 5100 - 6069 + 30 \bmod 26$$

$$= -939 \bmod 26$$

$$= -3 \bmod 26$$

$$= 23$$

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adjoint } K$$

To find Adjoint K

Repeat first 2 columns and then first 2 rows

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{matrix}$$

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{matrix}$$

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{matrix}$$

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \\ 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \end{matrix}$$

Now ignore first row and first column and use remanning 4x4 matrix for finding adjoint K

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \\ 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \end{matrix}$$

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \\ 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \end{matrix}$$

$$= \begin{array}{cccc} 18 & 21 & 21 & 18 \\ 2 & 19 & 2 & 2 \\ 17 & 5 & 17 & 17 \\ 18 & 21 & 21 & 18 \end{array}$$

$$= \begin{array}{c} 18\times19-2\times21 \end{array}$$

$$= \begin{array}{cccc} 18 & 21 & 21 & 18 \\ 2 & 19 & 2 & 2 \\ 17 & 5 & 17 & 17 \\ 18 & 21 & 21 & 18 \end{array}$$

$$- \quad 18\times19-2\times21 \qquad 2\times5-17\times19$$

$$= \begin{array}{cccc} 18 & 21 & 21 & 18 \\ 2 & 19 & 2 & 2 \\ 17 & 5 & 17 & 17 \\ 18 & 21 & 21 & 18 \end{array}$$

Performing the operation - Column wise

Entering the matrix - Row wise

$$= \quad 18\times19-2\times21 \qquad 2\times5-17\times19 \qquad 17\times21-18\times5$$

$$= \begin{array}{cccc} 18 & 21 & 21 & 18 \\ 2 & 19 & 2 & 2 \\ 17 & 5 & 17 & 17 \\ 18 & 21 & 21 & 18 \end{array}$$

Performing the operation - Column wise

Entering the matrix - Row wise

$$= \begin{array}{ccc} 18\text{x}19 - 2\text{x}21 & 2\text{x}5 - 17\text{x}19 & 17\text{x}21 - 18\text{x}5 \\ 21\text{x}2 - 19\text{x}21 & 19\text{x}17 - 5\text{x}2 & 5\text{x}21 - 21\text{x}17 \\ 21\text{x}2 - 2\text{x}18 & 2\text{x}17 - 17\text{x}2 & 17\text{x}18 - 21\text{x}17 \end{array}$$

$$= \begin{array}{ccc} 300 & -313 & 267 \\ -357 & 313 & -252 \\ 6 & 0 & -51 \end{array} \bmod 26$$

$$= \begin{array}{ccc} 14 & -1 & 7 \\ -19 & 1 & -18 \\ 6 & 0 & -25 \end{array} \bmod 26$$

$$= \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix}$$

Decryption requires K^{-1}, the inverse matrix K.

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

$$K^{-1} = \frac{1}{23} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

$$K^{-1} = \boxed{17} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \text{ mod } 26$$

$23^{-1} \times 23 = 1 \bmod 26$	$9 \times 23 = 25 \bmod 26$
$1 \times 23 = 23 \bmod 26$	$10 \times 23 = 22 \bmod 26$
$2 \times 23 = 20 \bmod 26$	$11 \times 23 = 19 \bmod 26$
$3 \times 23 = 17 \bmod 26$	$12 \times 23 = 16 \bmod 26$
$4 \times 23 = 14 \bmod 26$	$13 \times 23 = 13 \bmod 26$
$5 \times 23 = 11 \bmod 26$	$14 \times 23 = 10 \bmod 26$
$6 \times 23 = 8 \bmod 26$	$15 \times 23 = 7 \bmod 26$
$7 \times 23 = 5 \bmod 26$	$16 \times 23 = 4 \bmod 26$
$8 \times 23 = 2 \bmod 26$	$17 \times 23 = 1 \bmod 26$

$$K^{-1} = 17 \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

$$K^{-1} = \begin{pmatrix} 238 & 425 & 119 \\ 119 & 17 & 136 \\ 102 & 0 & 17 \end{pmatrix} \bmod 26$$

$$K^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

$$K \times K^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$K = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \quad K^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

This is demonstrated as

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} = \begin{pmatrix} 443 & 442 & 442 \\ 858 & 495 & 780 \\ 494 & 52 & 365 \end{pmatrix} \bmod 26$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Question: Decrypt "RRLMWBKASPDH" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Solution:

$P = C\,K^{-1} \bmod 26$

R	R	L	M	W	B	K	A	S	P	D	H
17	17	11	12	22	1	10	0	18	15	3	7

Decrypting: RRL

$$(P_1\ P_2\ P_3) = (R\ R\ L)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

Decryption

$$(C_1\ C_2\ C_3) = (17\ 17\ 14)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$= (17x4+17x15+11x24 \quad 17x9+17x17+11x0 \quad 17x15+17x6+11x17) \bmod 26$$

$$= (587\ 442\ 544) \bmod 26$$

$$= (15\ 0\ 24)$$

$$= (P\ A\ Y)$$

Decrypting: MWB

$$(P_1\ P_2\ P_3) = (M\ W\ B)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (12\ 22\ 1)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$= (12\text{x}4+22\text{x}15+1\text{x}24 \quad 12\text{x}9+22\text{x}17+1\text{x}0 \quad 12\text{x}15+22\text{x}6+1\text{x}17) \bmod 26$$

$$= (402\ 482\ 329) \bmod 26$$

$$= (12\ 14\ 17)$$

$$= (M\ O\ R)$$

Decrypting: KAS

$$(P_1\ P_2\ P_3) = (K\ A\ S)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (10\ 0\ 18)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$= (10\text{x}4+0\text{x}15+18\text{x}24 \quad 10\text{x}9+0\text{x}17+18\text{x}0 \quad 10\text{x}15+0\text{x}6+18\text{x}17) \bmod 26$$

$$= (472\ 90\ 456) \bmod 26$$

$$= (4\ 12\ 14)$$

$$= (E\ M\ O)$$

Decrypting: PDH

$$(P_1\ P_2\ P_3) = (P\ D\ H)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (15\ 3\ 7)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$= (15\text{x}4+3\text{x}15+7\text{x}24 \quad 15\text{x}9+3\text{x}17+7\text{x}0 \quad 15\text{x}15+3\text{x}6+7\text{x}17) \bmod 26$$

$$= (273\ 186\ 362) \bmod 26$$

$$= (13\ 4\ 24)$$

$$= (N\ E\ Y)$$

Question: Decrypt "RRLMWBKASPDH" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Ciphertext : RRLMWBKASPDH

Plaintext : pay more money

The Hill Algorithm

- This encryption algorithm takes *m* successive plaintext letters and substitutes them by *m* ciphertext letters. The substitution is determined by *m* linear equations in which each character is assigned a numerical value ($a = 0$, $b = 1, \ldots, z = 25$). For $m = 3$, the system can be described as

$$(c_1\ c_2\ c_3) = (p_1\ p_2\ p_3)\begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \bmod 26$$

In general terms, the Hill system can be expressed as

C = E(K , P) = P K mod 26

P = D(K , C) = C K^{-1} mod 26

$$\mathbf{K} = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Example: consider the plaintext “paymoremoney” and use the encryption key,

Solution: The first three letters of the plaintext are represented by the vector (15 0 24). Then, (15 0 24) K = (303 303 531) mod 26 = (17 17 11) = RRL. Continuing in this fashion, the ciphertext for the entire plaintext is RRLMWBKASPDH.